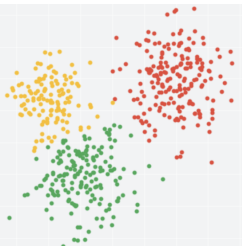


Divisive Clustering

Procheta Sen



Divisive (top-down) methods: main idea

- ▶ A top-down approach is used to successively partition the data objects into a tree-like structure.
- ▶ A flat clustering algorithm $\mathcal{A}$ (e.g. k-means algorithm) may be used for the partitioning in a given step. The algorithm $\mathcal{A}$ can be any arbitrary clustering algorithm, not necessarily a distance based algorithm.

Divisive (top-down) methods: main idea

- ▶ Provide flexibility in terms of choosing the trade-off between the balance in the tree structure and the balance in the number of objects in each cluster. For example:
- ▶ **Strategy 1:** split the heaviest node (a cluster with the maximum number of objects). Will result in leaf nodes (clusters) with a similar number of objects in them.
- ▶ **Strategy 2:** split each cluster into the same number of subclusters. Will result in a balanced tree structure with the same number of children at each node, but the leaf nodes (most granular clusters) will have varying numbers of objects.

Generic divisive clustering algorithm

Input: dataset: $\mathcal{D}$; flat algorithm: $\mathcal{A}$

1. **Initialise** tree $\mathcal{T}$ to contain a single (root) vertex with entire dataset $\mathcal{D}$
2. **Repeat:**
 - 2.1 Select a leaf node L in $\mathcal{T}$ based on pre-defined criterion;
 - 2.2 Use algorithm $\mathcal{A}$ split L into $L_1, \dots, L_k$;
 - 2.3 Add $L_1, \dots, L_k$ as children of L in $\mathcal{T}$;
3. **Until** termination criterion
4. **Return** current clustering or hierarchy or clusterings

Bisecting k -Means algorithm

Input: dataset: $\mathcal{D}$; number of clusters: s

1. **Initialise** tree $\mathcal{T}$ to contain a single (root) vertex with entire dataset $\mathcal{D}$

2. **Repeat:**

2.1 Select a leaf node (cluster) L in $\mathcal{T}$ that has the largest sum of square distance

$$\sum_{\bar{X}, \bar{Y} \in L} \text{dist}(\bar{X}, \bar{Y})^2;$$

2.2 Split L into 2 clusters L_1, L_2 using k -means algorithm;

2.3 Add L_1, L_2 as children of L in $\mathcal{T}$;

3. **Until** the number of leaf clusters is s

4. **Return** the leaf clusters

Bisecting k-Means algorithm: example

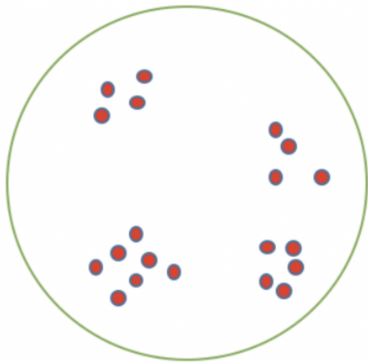


Figure:

Bisecting k-Means algorithm: example

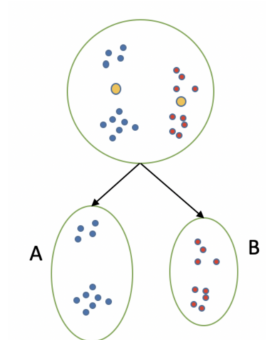


Figure:

Bisecting k-Means algorithm: example

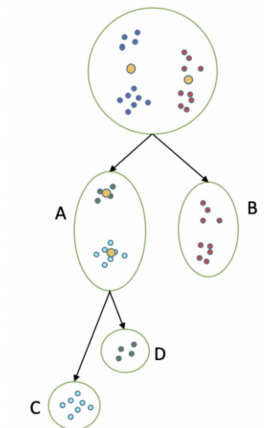


Figure:

Bisecting k-Means algorithm: example

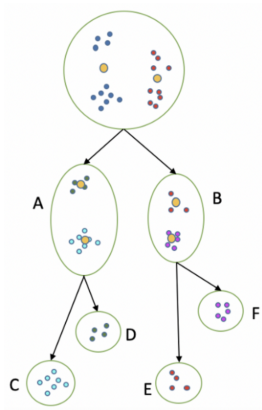


Figure: